<u>Group 1:</u>

Leah Colwell, [redacted]

Scott Morrison, [redacted]

Seb Smith, [redacted]

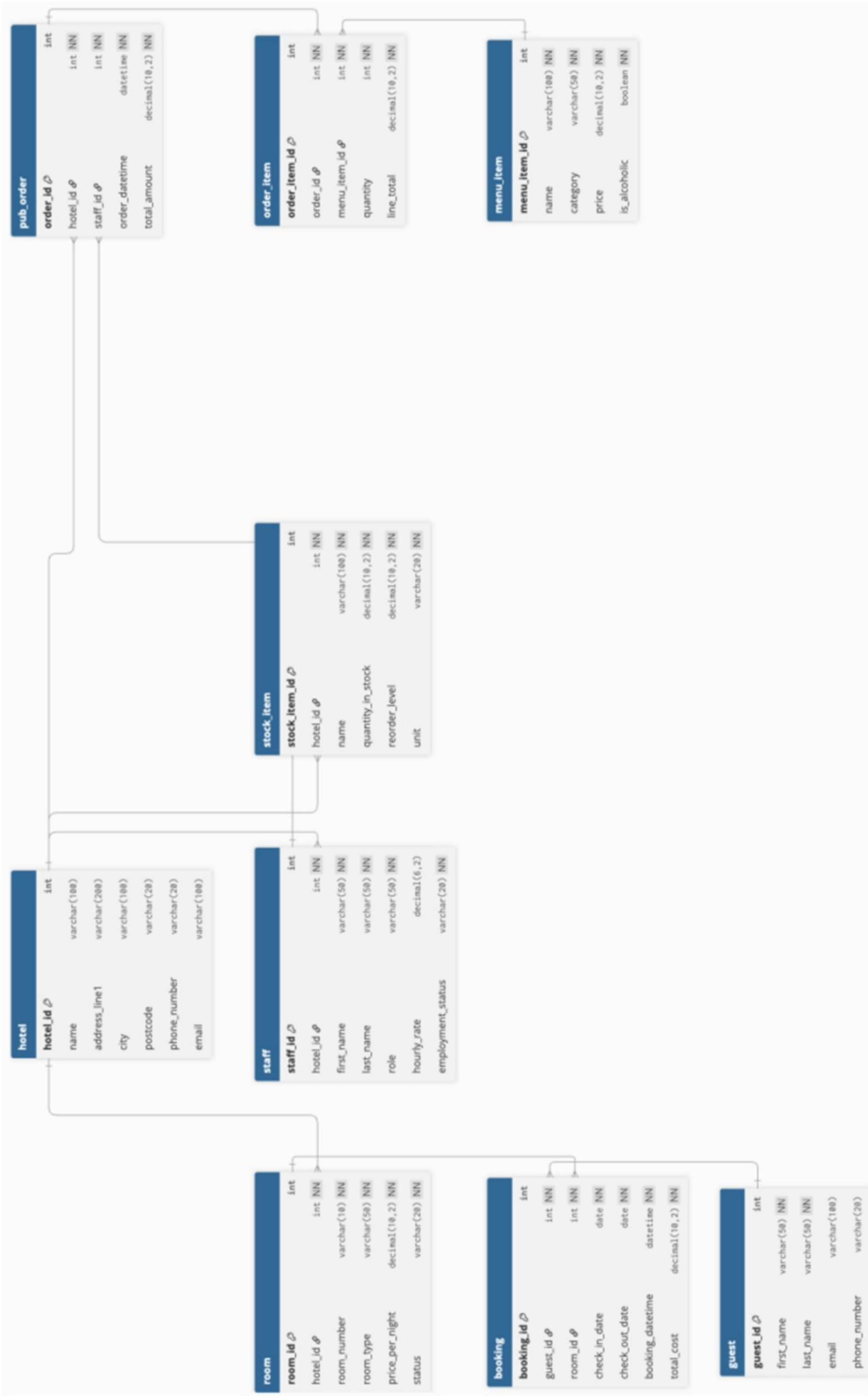
Emma Clark, [redacted]

Stephen Kelly, [redacted]

<u>Login details to access database in MySQL Workbench:</u>

- Username: [redacted]
- Password: [redacted]
- Connection name: [redacted]
- Host name: [redacted].[redacted]

<u>E-R diagram:</u>



Specification

The Crowned Duck Inn

Small independent Hotel/ Public house

The system models a small, independently operated hotel and public house offering 4 guest rooms for overnight accommodation alongside food and drink services. The business joins both short-stay visitors and local customers, which operates year-round. The Inn has developed a strong reputation for personalised service which attracts a loyal base of repeat customers.

The hotel employs a small team of staff that are responsible for managing bookings, preparing rooms, serving customers, and maintaining stock levels. As the business has grown, paper-based alongside informal digital records have become increasingly difficult to manage that has led to inefficiencies and a high-rise of errors in booking management as well as inventory tracking.

The purpose of the database system is to centralised and structure operational information. The system records details about the hotel premises, individual rooms, guests, bookings, staff members, and stock items used in the hotel's daily operations.

Each guest who stays at the hotel is recorded with relevant contact information. When a reservation is made, a booking is created, linking a guest to a specific room for defined check-in and check-out dates. This enables staff to accurately room allocation and reduces the risk of scheduling conflicts. Staff information is stored within the system, including roles, hourly rate, employment status, and supporting basic workforce administration.

As the business operates a pub serving food and beverage, stock records are maintained for supplies such as beverages, ingredients along with other consumables. Each stock item includes a quantity, unit of measurement along with a reorder threshold, which allows staff to monitor inventory levels to ensure continuity of supply. The database is intended to support management oversight and daily operational tests by providing consistent and reliable sources of information.

Database Design and Structure

The database was designed to reflect the core operational components of a small hotel and public house. The main entities represent physical resources, people, and administrative records required for daily management.

The Hotel entity represents the business premises and provides a central reference point for related operational data. It is linked to rooms, staff members, and stock items, which allows the system to support potential future expansion. The Room entity models individual guest rooms, storing information such as room number, type, nightly rate and availability status. Each room is associated with the hotel and may be linked to multiple bookings over time.

The Guest entity stores personal and contact details for customers, enabling the system to maintain accurate records that support repeat bookings. The Booking entity represents accommodation reservations that are linking guests to specific rooms for defined date ranges. This structure supports effective management of room occupancy and prevents scheduling conflicts.

The Staff entity stores information about employees, including their role, hourly rate, and employment status. This supports workforce administration and operational oversight. The Stock Item entity records inventory items used in food and beverage services. Attributes such as quantity, unit, and reorder level enable staff to monitor stock levels along with identifying when items need to be replenished.

The relationships between entities are implemented using foreign keys. For example, each booking refences one guest, one guest, while staff, stock items are linked to the hotel. This ensures data consistency along with reflecting real-world dependencies.

Business Rules and Design Decisions:

The structure of the database was informed by the operational requirements, practical constraints of a small hotel, and public house. Several business rules were identified during system analysis which are reflected in the final design.

Accommodation management was a central consideration. Each booking must be associated with exactly one guest and one room, reflecting the real-world process in which a room is reserved for a specific individual for a defined period. To support this rule, the Booking entity includes foreign keys referencing both the Guest and Room entities. Check-in and check-out dates are mandatory attributes, ensuring that the duration of each stay is clearly defined. Although a room may be used by many different guests over time, it should not be occupied by more than one guest during overlapping date ranges. This requirement supports operational checks for availability when new bookings are created.

Room attributes such as nightly rate and status are stored within the Room entity rather than within individual bookings. This avoids unnecessary duplication of data and ensures that changes to room pricing or availability are reflected consistently across the system.

Staff and inventory management were also key design considerations. Staff records include role, hourly rate and employment status to support workforce administration. Linking staff members to the Hotel entity reflects the organisational structure of the business and allows for clear management oversight. Stock items are recorded in quantities, units of measurement, and reorder level to support food and drink service. The inclusion of a reorder threshold reflects real-world practice, where supplies must be replenished before they run out. Storing this information centrally enables staff and managers to monitor inventory more effectively.

Data integrity was maintained through the use of primary and foreign key constraints. Primary keys ensure that each record can be uniquely identified, while foreign keys maintain valid relationships between entities. For example, a booking cannot exist without a valid guest and room; stock items cannot exist without being linked to the hotel. These constraints prevent incomplete or inconsistent data from being stored.

Overall, the database design reflects the operational needs of the business by supporting accommodation management, staff administration, and inventory control, while also enforcing realistic business rules and maintaining data consistency.

Implementation overview:

The database was implemented using MySQL and hosted on Amazon Web Services (AWS), enabling remote access through MySQL Workbench. The physical database structure was created using the SQL based on the logical model defined in the E-R diagram.

Tables were created using CREATE TABLE commands, with each entity implemented as a separate table. Primary keys were defined for all tables to ensure unique identification of records. Relationships between entities were established using ALTER TABLE commands to add foreign key constraints, ensuring referential integrity between related tables such as Booking, Guest, Room and Hotel.

Indexes were created on selected attributes to support efficient data retrieval. Appropriate data types were used for each attribute including integer types for identifiers, VARCHAR fields for textual information, DATE and DATETIME fields for booking and order records as well as DECIMAL fields for monetary values and stock quantities.

Sample data was inserted into each table using INSERT commands to test the structure and verify that relationships operated correctly. This test process confirmed that bookings, Staff records and Stock items could be stored and retrieved consistently.

The implemented database corresponds directly to the E-R diagram and logical design, ensuring consistency between conceptual planning and physical implementation.

Conclusion:

This report has described the development of a relational database system for a small hotel and public house. The system models the core operational components of the business, including accommodation management, guest records, staffing, stock control and food and beverage.

The database design reflects the real-world structure and daily functions of the business by separating key entities such as Guest, Room, Bookings, Staff, and Stock Items while maintaining appropriate relationships between them. Business rules relating to room allocation, booking periods, employment records, and inventory monitoring informed the logical structure of the system.

The implementation in MySQL, hosted on AWS, ensures that the conceptual design has been translated into a functional and structured database environment. Overall, the system provides a consistent and scalable foundation for managing the hotel and pubs' operations more efficiently than informal record-keeping methods.